Control of a string of vehicles

The system is depicted in the following figure.

Figure 10.1 String of vehicles

Each vehicle represents a "subsystem" of the overall plant and is characterized by two states:
- d_i: displacement with respect to the desired distance between vehicle i and its predecessor $i-1$ (if $i=1$, d_i is the vehicle's desired position);
- v_i: velocity of vehicle i.

The input u_i is the force imposed on vehicle i in horizontal direction and a friction force is exerted (i.e., with friction coefficient $h=1$ for all vehicles). For vehicle $i=1$, the dynamics is

$$\begin{cases} \dot{d}_1 = v_1 \\ \dot{v}_1 = -\dfrac{1}{m_1}v_1 + \dfrac{1}{m_1}u_1 \end{cases}$$

while, for a generic $i>1$,

$$\begin{cases} \dot{d}_i = v_i - v_{i-1} \\ \dot{v}_i = -\dfrac{1}{m_i}v_i + \dfrac{1}{m_i}u_i \end{cases}$$

Overall, considering $N=4$ vehicles, the system dynamics can be described by the centralized model

$$\dot{x} = Ax + Bu$$

given in the attached MATLAB file, where

$$x = \begin{bmatrix} p_1 \\ v_1 \\ \vdots \\ p_4 \\ v_4 \end{bmatrix}, u = \begin{bmatrix} u_1 \\ \vdots \\ u_4 \end{bmatrix}$$

and where it is assumed that all positions and velocities are measurable.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time $h = 0.1$ s). Perform the following analysis:
   a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
   b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions

a. Compute the continuous-time fixed modes
b. Compute the discrete-time fixed modes
c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the distances between the vehicles starting from a common random initial condition.